

LAB EXPERIMENT – 8

OUTPUTS:

1) Equi Join

```
MySQL 8.0 Command Line Cli x + v
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 28
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> use dbms;
Database changed
mysql> CREATE TABLE DEPARTMENT (
  -> DeptNo INT PRIMARY KEY,
  -> DeptName VARCHAR(20)
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO DEPARTMENT VALUES
  -> (1,'CSE'),
  -> (2,'ECE'),
  -> (3,'MECH'),
  -> (4,'CIVIL');
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> CREATE TABLE FACULTY (
  -> FacNo VARCHAR(4) PRIMARY KEY,
  -> FacName VARCHAR(20),
  -> DeptNo INT
  -> );
ERROR 1059 (42501): Table 'faculty' already exists
mysql> drop table faculty;
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE FACULTY (
  -> FacNo VARCHAR(4) PRIMARY KEY,
  -> FacName VARCHAR(20),
  -> DeptNo INT
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO FACULTY VALUES
  -> ('F001','Mohan',1),
  -> ('F002','Raju',2),
  -> ('F003','Ramesh',1),
  -> ('F004','Priya',3),
  -> ('F005','Kiran',4);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT *
  -> FROM FACULTY, DEPARTMENT
  -> WHERE FACULTY.DeptNo = DEPARTMENT.DeptNo;
+-----+
| FacNo | FacName | DeptNo | DeptNo | DeptName |
+-----+
| F001  | Mohan   | 1      | 1      | CSE      |
| F002  | Raju    | 2      | 2      | ECE      |
| F003  | Ramesh  | 1      | 1      | CSE      |
| F004  | Priya   | 3      | 3      | MECH     |
| F005  | Kiran   | 4      | 4      | CIVIL    |
+-----+
5 rows in set (0.00 sec)

mysql> |
```

2) Inner Join

```
0 rows in set (0.00 sec)

mysql> SELECT FACULTY.FacNo,
  -> FACULTY.FacName,
  -> DEPARTMENT.DeptName
  -> FROM FACULTY
  -> INNER JOIN DEPARTMENT
  -> ON FACULTY.DeptNo = DEPARTMENT.DeptNo;
+-----+
| FacNo | FacName | DeptName |
+-----+
| F001  | Mohan   | CSE      |
| F002  | Raju    | ECE      |
| F003  | Ramesh  | CSE      |
| F004  | Priya   | MECH     |
| F005  | Kiran   | CIVIL    |
+-----+
5 rows in set (0.00 sec)

mysql> |
```

3) Left Outer Join

```
mysql> SELECT FACULTY.FacNo,  
-> FACULTY.FacName,  
-> DEPARTMENT.DeptName  
-> FROM FACULTY  
-> LEFT JOIN DEPARTMENT  
-> ON FACULTY.DeptNo = DEPARTMENT.DeptNo;  
  
+-----+-----+-----+  
| FacNo | FacName | DeptName |  
+-----+-----+-----+  
| F001  | Mohan   | CSE      |  
| F002  | Raju    | ECE      |  
| F003  | Ramesh  | CSE      |  
| F004  | Priya   | MECH     |  
| F005  | Kiran   | CIVIL    |  
+-----+-----+-----+  
5 rows in set (0.00 sec)  
  
mysql> |
```

4) Right Outer Join

```
mysql> SELECT FACULTY.FacNo,  
-> FACULTY.FacName,  
-> DEPARTMENT.DeptName  
-> FROM FACULTY  
-> RIGHT JOIN DEPARTMENT  
-> ON FACULTY.DeptNo = DEPARTMENT.DeptNo;  
  
+-----+-----+-----+  
| FacNo | FacName | DeptName |  
+-----+-----+-----+  
| F003  | Ramesh  | CSE      |  
| F001  | Mohan   | CSE      |  
| F002  | Raju    | ECE      |  
| F004  | Priya   | MECH     |  
| F005  | Kiran   | CIVIL    |  
+-----+-----+-----+  
5 rows in set (0.00 sec)  
  
mysql> |
```

5) Full Outer Join

```
mysql> SELECT FACULTY.FacNo,  
-> FACULTY.FacName,  
-> DEPARTMENT.DeptName  
-> FROM FACULTY  
-> LEFT JOIN DEPARTMENT  
-> ON FACULTY.DeptNo = DEPARTMENT.DeptNo  
-> UNION  
-> SELECT FACULTY.FacNo,  
-> FACULTY.FacName,  
-> DEPARTMENT.DeptName  
-> FROM FACULTY  
-> RIGHT JOIN DEPARTMENT  
-> ON FACULTY.DeptNo = DEPARTMENT.DeptNo;  
  
+-----+-----+-----+  
| FacNo | FacName | DeptName |  
+-----+-----+-----+  
| F001  | Mohan   | CSE      |  
| F002  | Raju    | ECE      |  
| F003  | Ramesh  | CSE      |  
| F004  | Priya   | MECH     |  
| F005  | Kiran   | CIVIL    |  
+-----+-----+-----+  
5 rows in set (0.01 sec)  
  
mysql> |
```

6) Cross Join

```
mysql> SELECT FACULTY.FacNo,  
-> FACULTY.FacName,  
-> DEPARTMENT.DeptNo,  
-> DEPARTMENT.DeptName  
-> FROM FACULTY  
-> CROSS JOIN DEPARTMENT;
```

FacNo	FacName	DeptNo	DeptName
F001	Mohan	4	CIVIL
F001	Mohan	3	MECH
F001	Mohan	2	ECE
F001	Mohan	1	CSE
F002	Raju	4	CIVIL
F002	Raju	3	MECH
F002	Raju	2	ECE
F002	Raju	1	CSE
F003	Ramesh	4	CIVIL
F003	Ramesh	3	MECH
F003	Ramesh	2	ECE
F003	Ramesh	1	CSE
F004	Priya	4	CIVIL
F004	Priya	3	MECH
F004	Priya	2	ECE
F004	Priya	1	CSE
F005	Kiran	4	CIVIL
F005	Kiran	3	MECH
F005	Kiran	2	ECE
F005	Kiran	1	CSE

20 rows in set (0.00 sec)

7) Self Join

```
mysql> ALTER TABLE FACULTY  
-> ADD ManagerNo VARCHAR(4);  
Query OK, 0 rows affected (0.00 sec)  
Records: 0 Duplicates: 0 Warnings: 0  
  
mysql> UPDATE FACULTY SET ManagerNo='F001' WHERE FacNo='F002';  
Query OK, 1 row affected (0.01 sec)  
Rows matched: 1 Changed: 1 Warnings: 0  
  
mysql> UPDATE FACULTY SET ManagerNo='F001' WHERE FacNo='F003';  
Query OK, 1 row affected (0.01 sec)  
Rows matched: 1 Changed: 1 Warnings: 0  
  
mysql> UPDATE FACULTY SET ManagerNo='F002' WHERE FacNo='F004';  
Query OK, 1 row affected (0.01 sec)  
Rows matched: 1 Changed: 1 Warnings: 0  
  
mysql> UPDATE FACULTY SET ManagerNo='F002' WHERE FacNo='F005';  
Query OK, 1 row affected (0.01 sec)  
Rows matched: 1 Changed: 1 Warnings: 0  
  
mysql> SELECT  
-> F1.FacName AS Faculty,  
-> F2.FacName AS Manager  
-> FROM FACULTY F1  
-> LEFT JOIN FACULTY F2  
-> ON F1.ManagerNo = F2.FacNo;
```

Faculty	Manager
Mohan	NULL
Raju	Mohan
Ramesh	Mohan
Priya	Raju
Kiran	Raju

5 rows in set (0.00 sec)

```
mysql> |
```